Project: Worked examples
Title: Psychrometrics
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The state of moist air (humidity ratio, relative humidity, dew point, wet-bulb, enthalpy, specific volume, density) from dry-bulb and one other property at a stated pressure or altitude, and one air-handling process from that state with its heat and moisture balance: sensible heating or cooling, cooling with dehumidification, steam humidification, adiabatic humidification, or mixing with a second stream.
p_{ws} from Hyland–Wexler (over water ≥ 0 °C, over ice below); $W = 0.621945\, p_w/(p - p_w)$; $\varphi = p_w/p_{ws}$; $h = 1.006\, t + W(2501 + 1.86\, t)$ kJ/kg_da; $v = 0.287042\,(t + 273.15)(1 + 1.607858\, W)/p$; dew point and wet-bulb by bisection (wet-bulb from the adiabatic-saturation relation). Processes on a dry-air mass flow: sensible $Q = \dot{m}\,(h_2 - h_1)$ at constant W; cooling/dehumidification $Q_{total} = \dot{m}(h_1 - h_2) - \dot{m}_w\, h_w$ with condensate $\dot{m}_w = \dot{m}(W_1 - W_2)$; steam $h_2 = h_1 + (W_2 - W_1)\, h_{steam}$; adiabatic humidification at constant h; mixing by mass-weighted W and h.
Ref: ASHRAE Handbook — Fundamentals Ch. 1 — Psychrometrics (ideal-gas relations, processes)
Ref: Hyland & Wexler, ASHRAE Trans. 89(2A), 1983 — Saturation pressure of water and ice
Ref: CIBSE Guide C Section 1 — Properties of humid air (principles)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
altitude	Altitude (used when pressure is blank)	120	m	drawing: Site survey SRV-01: 120 m AOD	-500 … 5000 m
t_db	Dry-bulb temperature	22	degC	spec: Return air 22 °C / 40 %	-40 … 60 degC
known	Second known property	relative humidity		assumption: RH given.	any
RH	Relative humidity	40		spec: Return air 22 °C / 40 %	0 … 100
m_da	Dry-air mass flow	3	kg/s	spec: AHU-02 return 3 kg/s	0 … 1000 kg/s
process	Process	adiabatic mixing		assumption: Mixing box.	any
h_steam	Steam enthalpy (steam humidification)	2676		default	2500 … 3000
t_db_b	Second stream dry-bulb (mixing)	-3	degC	spec: CIBSE Guide A winter design −3 °C / 90 %, assumed	-40 … 60 degC
RH_b	Second stream relative humidity (mixing)	90		spec: Winter design condition	0 … 100
m_da_b	Second stream dry-air mass flow (mixing)	1	kg/s	spec: AHU-02 minimum outdoor air 1 kg/s	0 … 1000 kg/s

3. Assumptions

A1. Moist air is an ideal mixture of ideal gases; the enhancement factor is taken as 1. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A2. Condensate leaves at the coil leaving dry-bulb; steam adds enthalpy h_steam at negligible mass of dry air. (Process conventions.)
A3. Adiabatic humidification follows a line of constant enthalpy (close to constant wet-bulb). (Evaporative process idealisation.)
A4. known (Second known property) is taken as relative humidity. (RH given.)
A5. process (Process) is taken as adiabatic mixing. (Mixing box.)

4. Working

```
[2] p = 101.325 (1 - 2.25577e-5 z)^5.2559                    ASHRAE Ch. 1 eq. 3
= z = 120 m
= 99.892 kPa
```
Barometric pressure (standard atmosphere)
```
[6] p_ws_1 = p_ws(t) Hyland-Wexler (entering)
= t = 22 °C
= 2.6448 kPa
```
Saturation pressure, state 1
```
[7] p_w_1 = p W/(0.621945 + W)
= W = 0.0066572
= 1.0579 kPa
```
Vapour pressure, state 1
```
[8] t_1_out = dry-bulb (entering)
=
= 22 degC
```
Dry-bulb, state 1
```
[9] W_1 = 0.621945 p_w/(p - p_w)
= 1.0579 / (99.892 - 1.0579)
= 6.6572
```
Humidity ratio, state 1 (g/kg dry air)
```
[10] RH_1 = p_w / p_ws
= 1.0579 / 2.6448
= 40
```
Relative humidity, state 1 (%)
```
[11] phi_1 = p_w / p_ws
= 1.0579 / 2.6448
= 40 %
```
Relative humidity, state 1, as a percentage quantity (the same value as RH_1; a plain-number output cannot be linked to a percentage input)
```
[12] t_dp_1 = p_ws(t_dp) = p_w, bisection
= p_w = 1.0579 kPa
= 7.7942 degC
```
Dew point, state 1
```
[13] t_wb_1 = adiabatic-saturation relation, bisection
= t = 22, W = 0.0066572
= 13.853 degC
```
Wet-bulb, state 1
```
[14] h_1 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 22 + 0.0066572 × (2501 + 1.86 × 22)
= 39.054
```
Enthalpy, state 1 (kJ/kg dry air)
```
[15] v_1 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 99.892 kPa
= 0.8572 m³/kg
```

Specific volume, state 1 (per kg dry air)

[16] rho_1 = (1 + W)/v

=

= 1.1744 kg/m³

Moist-air density, state 1

[22] p_ws_b = p_ws(t) Hyland-Wexler (second stream)

= t = -3 °C

= 0.47606 kPa

Saturation pressure, state b

[23] p_w_b = p W/(0.621945 + W)

= W = 0.0026791

= 0.42845 kPa

Vapour pressure, state b

[24] t_b_out = dry-bulb (second stream)

=

= -3 degC

Dry-bulb, state b

[25] W_b = 0.621945 p_w/(p - p_w)

= 0.42845 / (99.892 - 0.42845)

= 2.6791

Humidity ratio, state b (g/kg dry air)

[26] RH_b = p_w / p_ws

= 0.42845 / 0.47606

= 90

Relative humidity, state b (%)

[27] phi_b = p_w / p_ws

= 0.42845 / 0.47606

= 90 %

Relative humidity, state b, as a percentage quantity (the same value as RH_b; a plain-number output cannot be linked to a percentage input)

[28] t_dp_b = p_ws(t_dp) = p_w, bisection

= p_w = 0.42845 kPa

= -4.2454 degC

Dew point, state b

[29] t_wb_b = adiabatic-saturation relation, bisection

= t = -3, W = 0.0026791

= -3.4949 degC

Wet-bulb, state b

[30] h_b = 1.006 t + W (2501 + 1.86 t)

= 1.006 × -3 + 0.0026791 × (2501 + 1.86 × -3)

= 3.6675

Enthalpy, state b (kJ/kg dry air)

[31] v_b = 0.287042 (t + 273.15)(1 + 1.607858 W)/p

= p = 99.892 kPa

= 0.77963 m³/kg

Specific volume, state b (per kg dry air)

[32] rho_b = (1 + W)/v

=

= 1.2861 kg/m³

Moist-air density, state b

[33] m_da_3 = □_a + □_b

= 3 + 1

= 4 kg/s

Mixed dry-air mass flow

[34] p_ws_2 = p_ws(t) Hyland-Wexler (mixed)
= t = 15.784 °C
= 1.7935 kPa
Saturation pressure, state 2
[35] p_w_2 = p W/(0.621945 + W)
= W = 0.0056627
= 0.90129 kPa
Vapour pressure, state 2
[36] t_2_out = dry-bulb (mixed)
=
= 15.784 degC
Dry-bulb, state 2
[37] W_2 = 0.621945 p_w/(p - p_w)
= 0.90129 / (99.892 - 0.90129)
= 5.6627
Humidity ratio, state 2 (g/kg dry air)
[38] RH_2 = p_w / p_ws
= 0.90129 / 1.7935
= 50.253
Relative humidity, state 2 (%)
[39] phi_2 = p_w / p_ws
= 0.90129 / 1.7935
= 50.253 %
Relative humidity, state 2, as a percentage quantity (the same value as RH_2; a plain-number output cannot be linked to a percentage input)
[40] t_dp_2 = p_ws(t_dp) = p_w, bisection
= p_w = 0.90129 kPa
= 5.4665 degC
Dew point, state 2
[41] t_wb_2 = adiabatic-saturation relation, bisection
= t = 15.784, W = 0.0056627
= 10.314 degC
Wet-bulb, state 2
[42] h_2 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 15.784 + 0.0056627 × (2501 + 1.86 × 15.784)
= 30.207
Enthalpy, state 2 (kJ/kg dry air)
[43] v_2 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 99.892 kPa
= 0.83782 m³/kg
Specific volume, state 2 (per kg dry air)
[44] rho_2 = (1 + W)/v
=
= 1.2003 kg/m³
Moist-air density, state 2

5. Results

SYMBOL	VALUE	DESCRIPTION
t_1_out	22 degC	Dry-bulb 1
p_ws_1	2.6448 kPa	p_ws 1
p_w_1	1.0579 kPa	p_w 1
W_1	6.6572	W 1 (g/kg)

SYMBOL	VALUE	DESCRIPTION
RH_1	40	RH 1 (%)
phi_1	40 %	φ 1 (RH as a % quantity, for linking)
t_dp_1	7.7942 degC	Dew point 1
t_wb_1	13.853 degC	Wet-bulb 1
h_1	39.054	h 1 (kJ/kg)
v_1	0.8572 m³/kg	v 1
rho_1	1.1744 kg/m³	ρ 1
t_b_out	-3 degC	Dry-bulb b
p_ws_b	0.47606 kPa	p_ws b
p_w_b	0.42845 kPa	p_w b
W_b	2.6791	W b (g/kg)
RH_b	90	RH b (%)
phi_b	90 %	φ b (RH as a % quantity, for linking)
t_dp_b	-4.2454 degC	Dew point b
t_wb_b	-3.4949 degC	Wet-bulb b
h_b	3.6675	h b (kJ/kg)
v_b	0.77963 m³/kg	v b
rho_b	1.2861 kg/m³	ρ b
t_2_out	15.784 degC	Dry-bulb 2
p_ws_2	1.7935 kPa	p_ws 2
p_w_2	0.90129 kPa	p_w 2
W_2	5.6627	W 2 (g/kg)
RH_2	50.253	RH 2 (%)
phi_2	50.253 %	φ 2 (RH as a % quantity, for linking)
t_dp_2	5.4665 degC	Dew point 2
t_wb_2	10.314 degC	Wet-bulb 2
h_2	30.207	h 2 (kJ/kg)
v_2	0.83782 m³/kg	v 2
rho_2	1.2003 kg/m³	ρ 2
m_da_3	4 kg/s	Mixed mass flow

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

WARNING — input.h_steam: h_steam (Steam enthalpy (steam humidification)) was not entered; the template default 2676 is used.

8. Verification

Template bs.psychrometrics@1.1.0 Content hash 019c21a8f58b1d42 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0